# B3.1 Gas exchange (Plant)

*Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-b3-1-plant.html*

*Written for this site from the course textbook and class slides. Use the page numbers to read the book's fuller account. Gas exchange in animals (B3.1.1–6) is in Body Systems. Image credits are on the last slide of the slides for this subtopic.*

## B3.1.7 How leaves are adapted to exchange gases

pp. 283–286

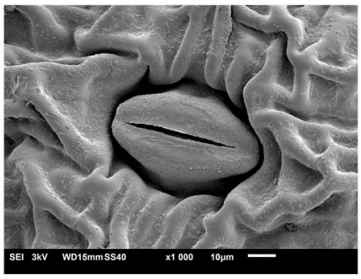
A leaf must take in carbon dioxide for photosynthesis, while limiting its loss of water. Being flat and thin gives it a large surface area and short diffusion distances.

| STRUCTURE | ADAPTATION |
|---|---|
| WAXY CUTICLE | A largely impermeable layer, so gases and water vapour must pass through stomata, where flow is controlled |
| EPIDERMIS | Most pores are in the lower epidermis, which is shaded and cooler, and this reduces water loss |
| AIR SPACES | Let gases circulate between loosely packed cells, maintaining gradients |
| SPONGY MESOPHYLL | A large, moist surface for gases to enter and leave cells |
| STOMATAL GUARD CELLS | Open and close the pore, controlling gas exchange and water loss |
| VEINS | Xylem supplies water, and phloem removes sugars, which helps to maintain gradients |

When guard cells take up water they become **turgid** and curve apart, opening the pore. When they lose water they become **flaccid** and the pore closes.

### Guard cells control the stoma

B3.1.7




Stoma and guard cells. Judyta Dulnik, CC BY-SA 4.0.




Water enters the guard cells, they become **turgid** and curve apart: the pore **opens**. When they lose water they become **flaccid** and it **closes**.

**Common mistake**
The stoma is the pore; guard cells are the two cells around it.



**Watch out:** the stoma is the pore, and the guard cells are the cells around it.

### B3.1.8 How tissues are arranged in a leaf

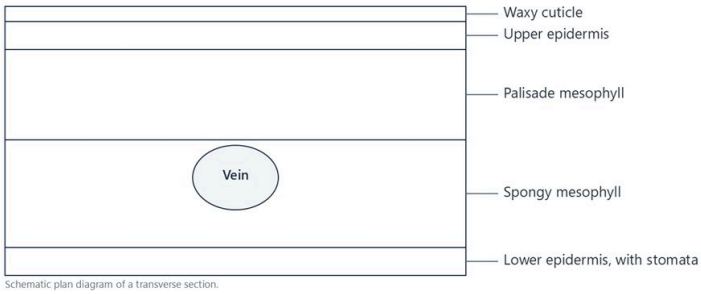
pp. 284–286

From top to bottom, a dicot leaf has a waxy cuticle, upper epidermis, palisade mesophyll, spongy mesophyll and lower epidermis with stomata. Veins, containing xylem and phloem, run through the mesophyll.

A **plan diagram** shows these tissues as regions. Use clear, continuous lines, do not shade, and do not draw individual cells.

#### Plan diagram of a dicot leaf

B3.1.8




Schematic plan diagram of a transverse section.

**Plan diagram**
Show tissues as regions. No individual cells, no shading.

**Activity**
Draw a plan diagram from a micrograph of a leaf section. Use continuous lines, keep the proportions, and label cuticle, epidermis, palisade and spongy mesophyll, xylem and phloem.

B3.1 Gas exchange in plants – Textbook pp. 284–286 8

## LOSING WATER

### B3.1.9 Why gas exchange in a leaf leads to transpiration

pp. 286–288

For carbon dioxide to enter, stomata must be open. Water evaporates from moist cell walls into the air spaces, and water vapour diffuses out through the same pores. This loss is **transpiration**.

| FACTOR | EFFECT OF AN INCREASE | WHY |
|---|---|---|
| TEMPERATURE | Rate rises | More evaporation, so a greater concentration difference of water vapour |
| HUMIDITY | Rate falls | The air outside holds more water vapour, so the gradient is smaller |
| WIND | Rate rises | Water vapour near the stomata is swept away, keeping the gradient steep |
| LIGHT | Rate rises | Stomata open for photosynthesis, and the leaf warms |

A **potometer** measures the rate at which a shoot takes up water, using the movement of an air bubble along a capillary tube. A small amount of this water stays in the plant, so it is an indirect measure.

**Watch out:** water vapour diffuses out, not liquid water, and this step is not osmosis.

### B3.1.10 How many stomata a leaf has

pp. 289–291

**Stomatal density** is the number of stomata per unit area of leaf. More stomata can take up more carbon dioxide, but they can also lose more water.

1. Paint clear nail varnish on the underside of a leaf, and let it dry.
2. Lift it off with clear tape, and stick it to a slide.
3. Count the stomata in several fields of view, and find the mean.
4. Area of field = 3.14 × $r^2$. Density = mean count ÷ area.

© 2026 Akash Parthi Bendran. Please ask before using or sharing: see whatthecell.org/about.html#permission